HEM SHEELA MODEL SCHOOL

DURGAPUR

FINAL TERM EXAMINATION (2025-2026)

SUBJECT - MATHEMATICS

CLASS-VI

Time allowed: 03 Hours. *Date: 26/02/2026* *Total Marks: 80*

General Instructions:
1. All questions are compulsory.
2. Section A consists of 20 questions carrying 1 mark each.
3. Section B consists of 5 questions carrying 2 marks each.
4. Section C consists of 6 questions carrying 3 marks each.
5. Section D consists of 8 questions carrying 4 marks each.

SECTION A

(Choose the correct option)

1. Find the 10th term in the sequence of Virahanka Numbers:- 1
   a) 45 b) 50 c) 55 d) 34
2. A cricket ground is in the shape of a quadrilateral. If a batsman hits a four, then the ball has to touch a point :- 1
   a) in the exterior of the quadrilateral b) on the quadrilateral
   c) in the interior of the quadrilateral d) none of these
3. The cost of putting a fence around a square field at Rs. 2.50 per m is Rs. 200.The length of each side of the field is:- 1
   a) 80 m b) 40 m c) 20 m d) None
4. If $\frac{1}{2}+\frac{1}{x}=2$ then the value of x is:- 1
   a) $\frac{3}{2}$ b) $-\frac{2}{3}$ c) $\frac{2}{3}$ d) $-\frac{3}{2}$
5. $4\,m+35\,cm+410\,cm$ in meters is equal to:- 1
   a) 80 m b) 8.45 m c) 845 m d) 0.845 m
6. If 4 , a , a , 36 are in proportion, then the value of a is: 1
   a) 24 b) 12 c) 3 d) 124
7. Each numerical figure in a data is called an _____. 1
   a) Raw data b) Frequency c) Observation d) Array
8. The perimeter of a square with area $2500m^2$ is:- 1
   a) 2000 m b) 20 m c) 200 m d) 2m
9. The sum of 100 counting numbers adding up and down is equal to:- 1
   a) 1000 b) 10000 c) 10 d) 100000
10. Total number of diameters of a circle are:- 1
    a) 1 b) 2 c) 4 d) uncountable numbers
11. When 0.02 is written as a fraction in the simplest form, the sum of the numerator and denominator is: 1
    a) 51 b) 21 c) 12 d) 100
12. An electric pole casts a shadow of length 20 m at a time when a tree of 6 m high casts a shadow of length 8 m. The height of the pole is :- 1
    a) $\frac{80}{3}$ b) 15 c) 12 d) $\frac{3}{80}$
13. A triangle whose one angle is more than 90° is called:- 1
    a) a Scalene Triangle b) a Right Triangle
    c) an Acute Triangle d) an Obtuse Triangle
14. The value of $\frac{5}{6}+\frac{2}{3}-\frac{4}{9}$ is :- 1
    a) $1\frac{1}{3}$ b) $1\frac{1}{6}$ c) $1\frac{1}{9}$ d) $1\frac{1}{18}$

15. The heights of Jaya, Sabina, Sanju and Rohit are 1.61m, 1.64 m, 1.60 m and 1.62 m respectively. Who is the tallest of them all ? 1

a) Sabina b) Jaya c) Sanju d) Rohit

16. One of the acute angles of a right triangle is 60°, find the other acute angle. 1

a) 50° b) 30° c) 120° d) 40°

17. In the word MATHEMATICS, the ratio of the number of consonants to the number of vowels is: 1

a) 4:7 b) 7:4 c) 5:6 d) 6:5

18. The figure bounded by an arc and the two radii joining the end points of the arc with the centre is called a _____ of the circle. 1

a) Sector b) Segment c) Central Angle d) Semi circle

DIRECTION: In the question number 19 and 20, a statement of assertion (A) is followed by a statement of Reason (R). Choose the correct option

(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A)
(b) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A)
(c) Assertion (A) is true but reason (R) is false.
(d) Assertion (A) is false but reason (R) is true

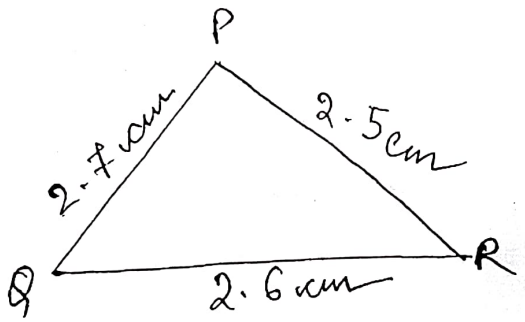
19. **Assertion (A) :-** A quadrilateral has 4 angles and 4 diagonals. 1
Reason (R):- A quadrilateral is a polygon made up of line segments.

20. **Assertion (A):-** The circumference of a circle of radius r is equal to $2\pi r$. 1
Reason (R):- the ratio of the circumference and the diameter of a circle is equal to π.

SECTION B

21. Ankita is 18 years old and her brother Apoorv is 16 years old. What will be the ratio of Apoorv's age to Ankita's age after 8 years? 2

22. 2



1) Name the triangle PQR.
2) Mention the vertex opposite to side QR in $\Delta\, PQR$.

23. The weight of a cylinder filled with gas is $30kg$. If the weight of the empty cylinder is $14kg\ 80g$, find the weight of the gas contained in it. 2

24. Draw a line AB. Take a point P on it. Draw a line passing through P and perpendicular to AB. 2

25. How many square tiles each of side 0.5 m will be required to pave the floor of a room which is 4 m long and 3 m broad? 2

SECTION C

26. Draw a neat diagram of a Rhombus and write any two properties of it. 3

27. In a ΔABC, if $3\angle A = 4\angle B = 6\angle C$. Calculate the smallest angle. 3

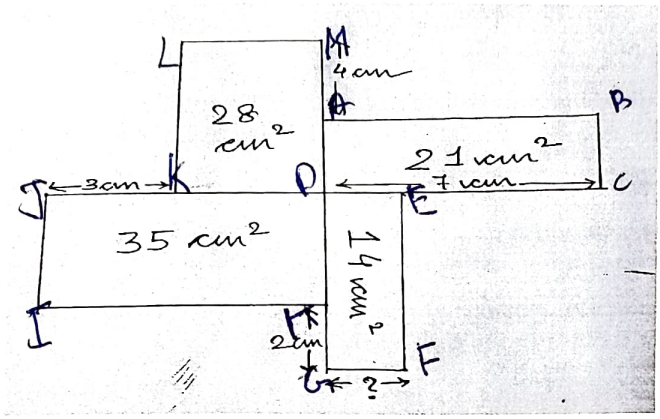
P.T.O

13, 9, 6
11, 4, 2.

28. The data below shows the ages of people at a party 35, 50, 37, 44, 53, 41, 39, 45, 48, 56. Draw an ordered stem and leaf diagram for the data. 3

29. A wheel barrow can hold $26\frac{1}{4}kg$. Four rocks that weigh $6\frac{1}{8}kg$, $8\frac{1}{2}kg$, $4\frac{3}{4}\ kg$ and $7\frac{1}{2}\ kg$ are to be loaded into the wheel barrow. Can the wheel barrow hold all the four rocks? 3

30. Find the missing value of the length of a side in the given figure. 3



31. If we begin the sequence of the powers of 2 with 1 as the first term, then what will be the 8th term of this sequence? Find the sum of the first seven terms of this sequence. 3

SECTION D

32. Draw a circle of radius 6cm. In the circle, draw a chord AB 4
i) If O is the centre of the circle, join OA and OB.
ii) Measure the chord AB
iii) Shade the major sector of the circle.

33. The ratio of the speeds of two vehicles is 2: 3. If the second vehicle covers $72km$ in 2 hours, what is the speed of the first vehicle? 4

34. Ram Babu collected data on the number of tickets sold at the Inter State Bus Terminal in Delhi for a few different cities over a 2-hour period. 4

City	Ajmer	Kota	Jaipur	Udaipur	Ahoar
No. of tickets	24	20	16	28	16

Help his assistant Brij to use this data and prepare a bar graph .

35. A can hold $12.5l$ of mixed fruit juice. $4.035l$ of apple juice and $6l\ 15ml$ of orange juice have been poured in the can. What would be the amount of grape juice that can still be added in the can? 4

36. Construct a rectangle one of whose sides is 4 cm and the diagonal is of length 8 cm. 4

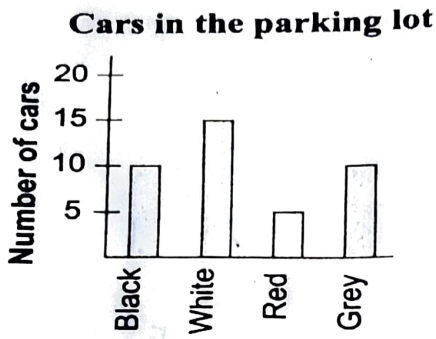
37. Find the perimeter of a rectangular field whose length is four times its width and which has an area equal to 324 cm^2. 4

CASE STUDIES

38. A boy is cycling in such a way that the wheels of his bicycle are making 140 revolutions in 1 hour. If the diameter of the wheel is 350 cm. 4 (1+2+1)

1) Find the circumference of the wheel.
2) Find the distance (in Km) the boy will cover.
3) Find the speed (in km/hr) at which the boy is cycling.

39. Read the bar graph which shows the number of cars in the parking lot and answer the following questions: 4



1) Which colour is the least colour common among the cars in the parking lot?
2) Which colours had the same number of cars parked?
3) What is the ratio of the number of white cars to red cars?
4) Find the total number of cars in the parking lot.